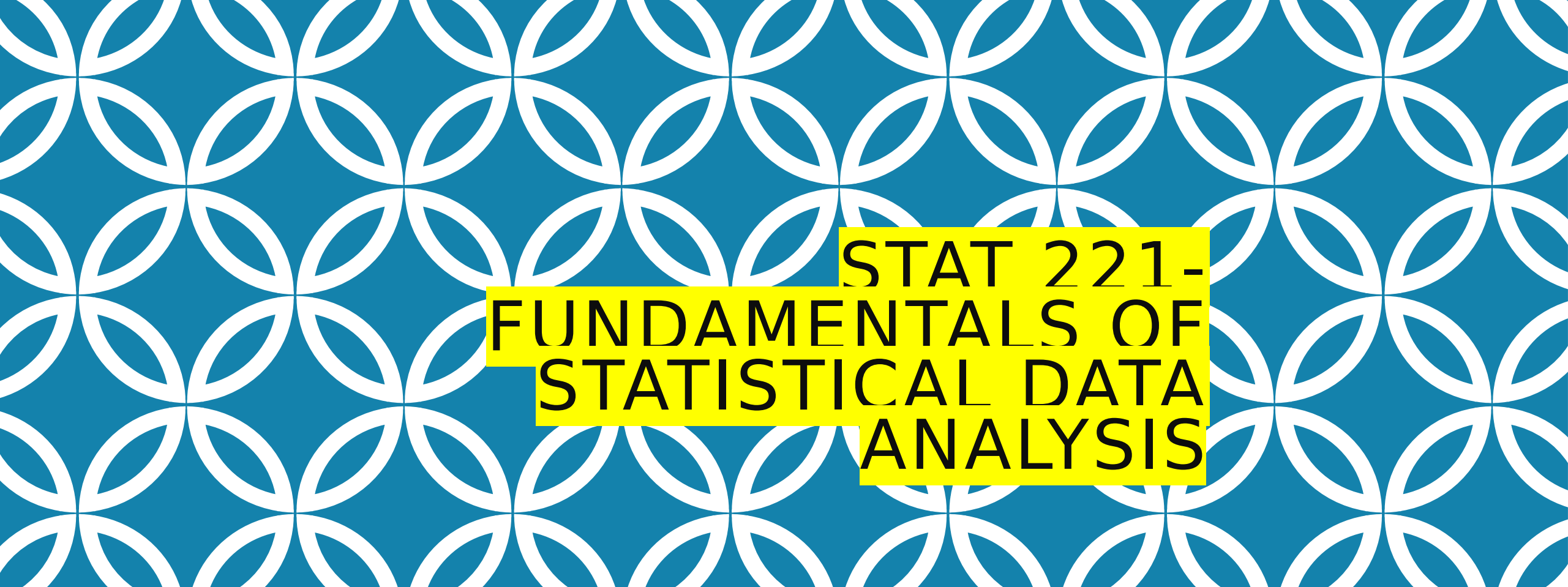
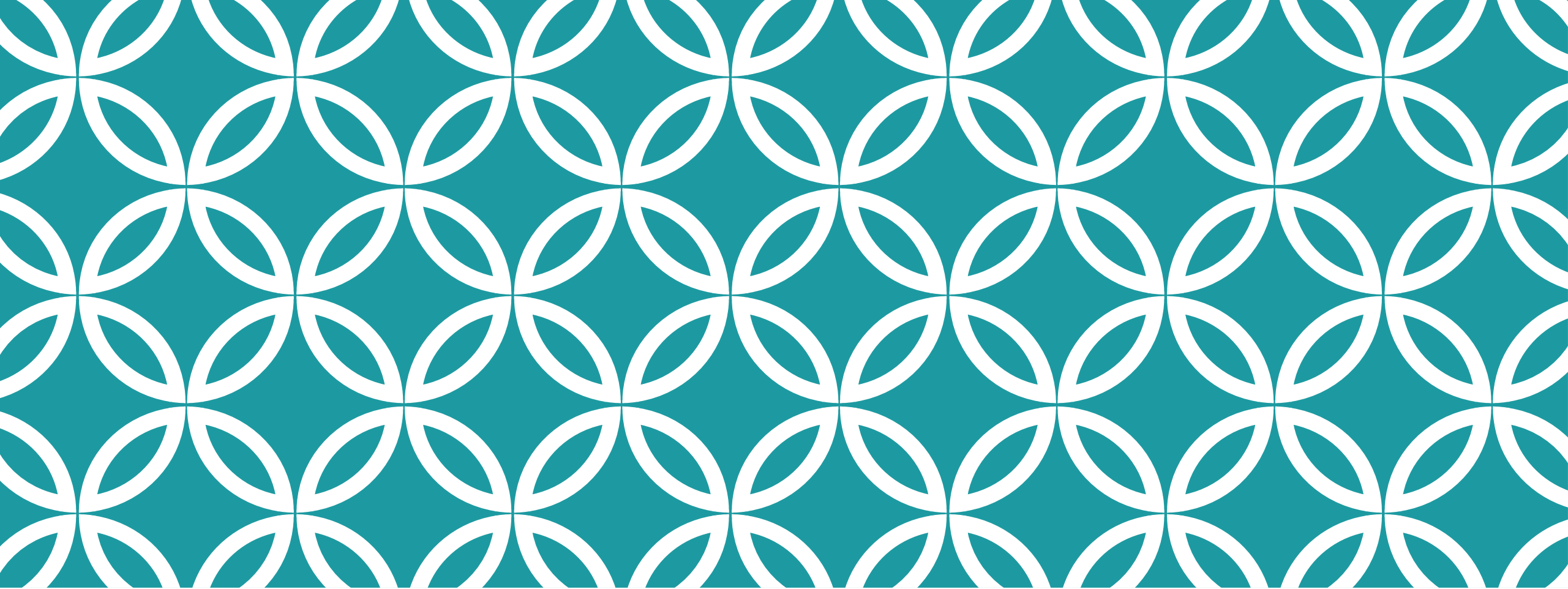




STAT 221- FUNDAMENTALS OF STATISTICAL DATA ANALYSIS

Week-13: Hypothesis Testing

Shivakumar Jolad

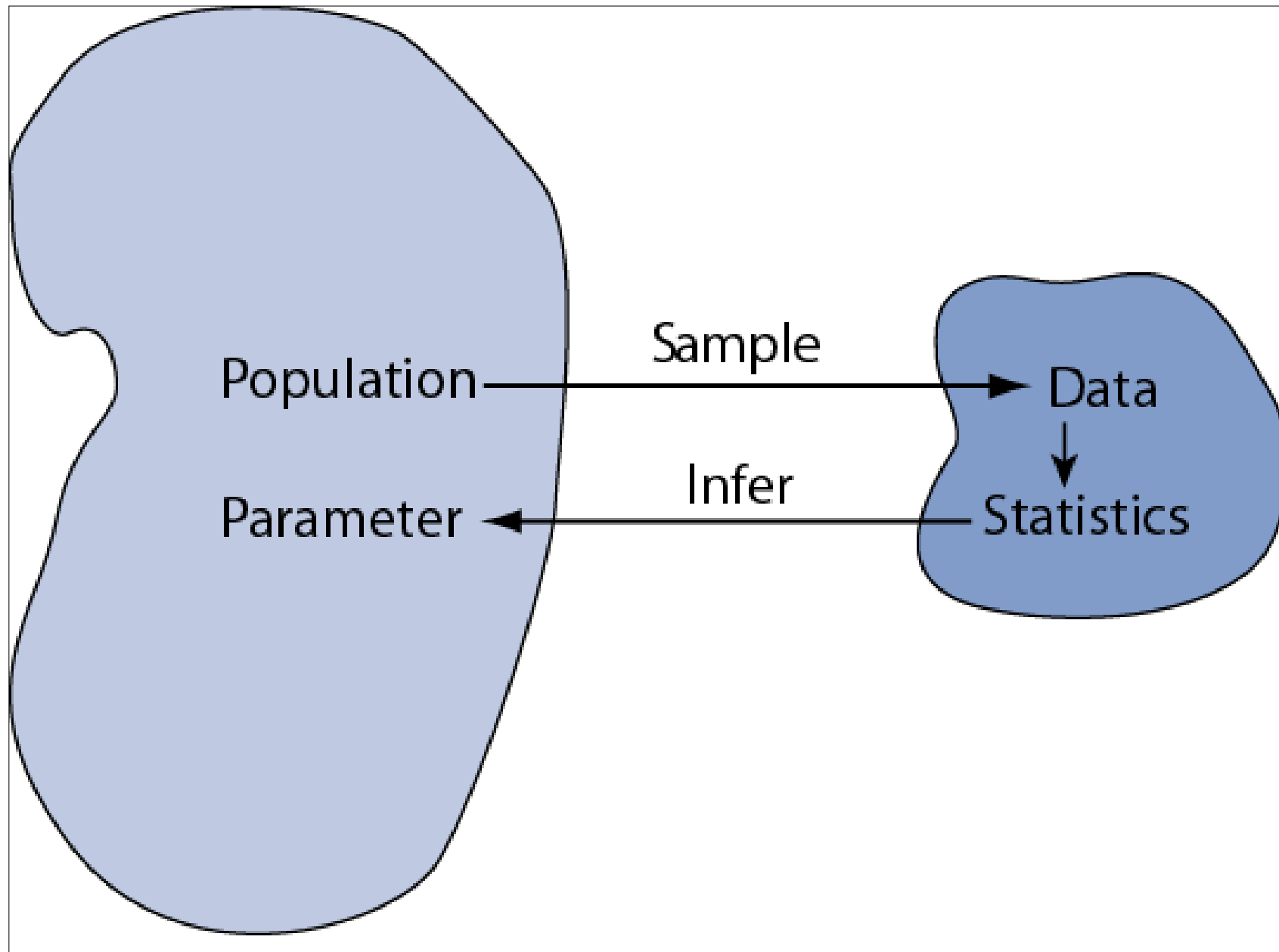


HYPOTHESIS TESTING



DISTINCTIONS BETWEEN PARAMETERS AND STATISTICS (CHAPTER 8 REVIEW)

	Parameters	Statistics
Source	Population	Sample
Notation	Greek (e.g., μ)	Roman (e.g., $\bar{x}$)
Vary	No	Yes
Calculated	No	Yes

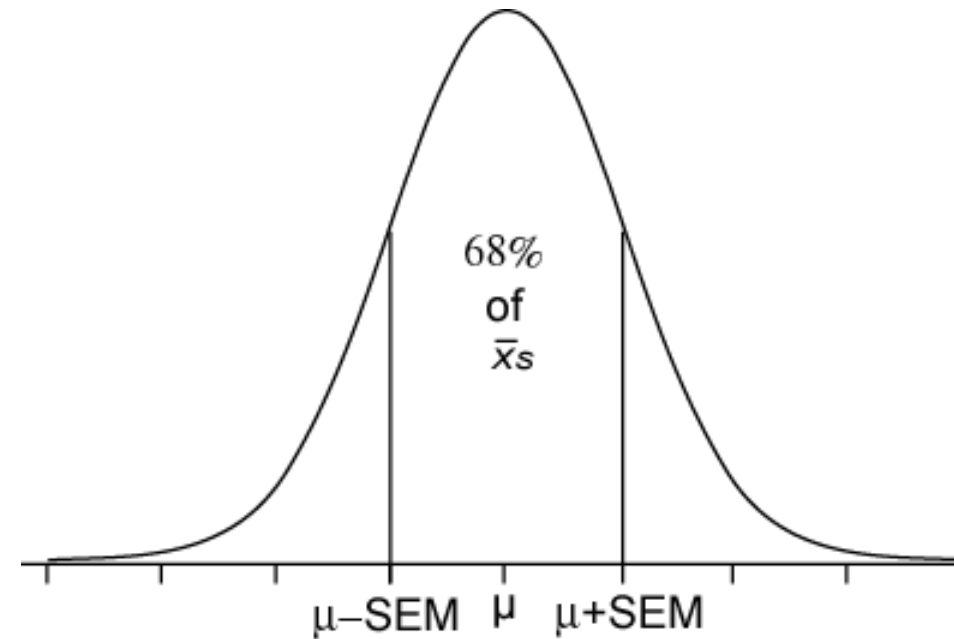


SAMPLING DISTRIBUTIONS OF A MEAN (LAST CLASS)

The **sampling distributions of a mean (SDM)** describes the behavior of a sampling mean

$$\bar{x} \sim N(\mu, SE_{\bar{x}})$$

$$\text{where } SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$



Three important sampling postulates: the central limit theorem, the law of large numbers (unbiased nature of the sample mean), and square root law.

WATCH THE VIDEOS

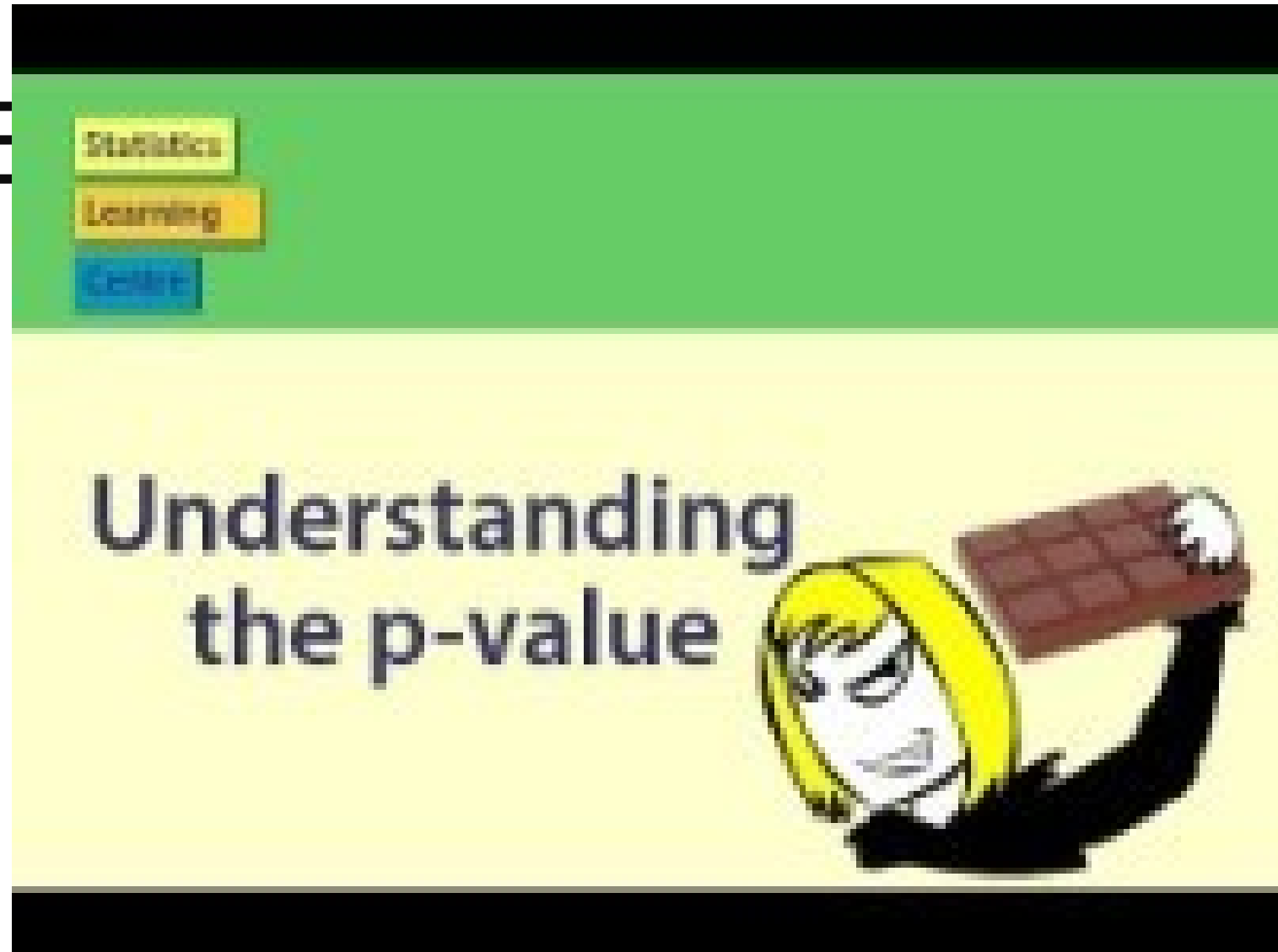
Basics (for your revision)



Understanding Statistical Inference: 6:30 mins

<https://www.youtube.com/watch?v=tFRXsngz4UQ&list=PLm9FYjKtq7Pzjh7e727hSr8VvSR9OvqsZ>

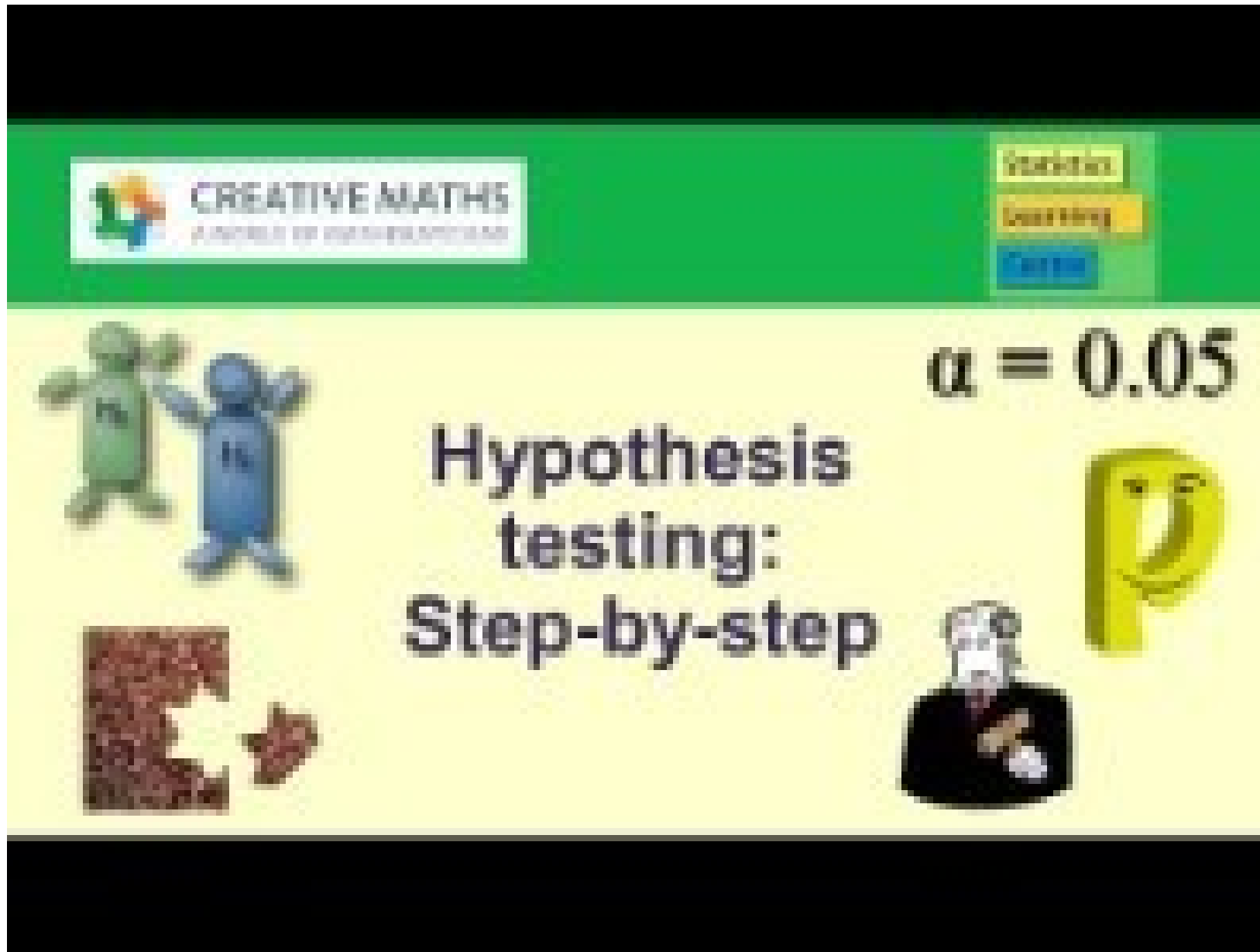
P-VALUE



Understanding p-value (4:30 mins)

[https://www.youtube.com/watch?](https://www.youtube.com/watch?v=eyknGvncKLw)

[v=eyknGvncKLw&list=PLm9FYjKtq7Pzjh7e727hSr8VvSR9OvqsZ&index=2](https://www.youtube.com/watch?v=eyknGvncKLw&list=PLm9FYjKtq7Pzjh7e727hSr8VvSR9OvqsZ&index=2)



Hypothesis Testing (7:30 mins): <https://www.youtube.com/watch?v=0zZYBALbZgg&list=PLm9FYjKtq7Pzjh7e727hSr8VvSR9OvqsZ&index=3>

Hypothesis Testing

Two-tailed

$$H_0: \mu = 23$$

$$H_1: \mu \neq 23$$



One-tailed

Left-tailed

$$H_0: \mu \geq 23$$

$$H_1: \mu < 23$$



Right-tailed

$$H_0: \mu \leq 23$$

$$H_1: \mu > 23$$



Hypothesis testing (4 mins):

<https://www.youtube.com/watch?v=DlwOTOydeyk>

HYPOTHESIS TESTING

Is also called ***significance testing***

Tests a claim about a parameter using evidence (data in a sample)

The technique is introduced by considering **a one-sample z test**

Procedure:

- Null and alternative hypotheses
- Test statistic
- P-value and interpretation
- Significance level (optional)

NULL AND ALTERNATIVE HYPOTHESES

Convert the research question to null and alternative hypotheses

The **null hypothesis (H_0)** is a claim of “no difference in the population”

The **alternative hypothesis (H_a)** claims “ H_0 is false”

Collect data and seek evidence against H_0 as a way of bolstering H_a (deduction)

ILLUSTRATIVE EXAMPLE: “BODY WEIGHT”

The problem: In the 1970s, 20–29 year old men in the U.S. had a **mean μ body weight of 170 pounds.** **Standard deviation σ was 40 pounds.** We test whether mean body weight in the population now differs.

Null hypothesis $H_0: \mu = 170$ (“no difference”)

The **alternative hypothesis** can be either

$H_a: \mu > 170$ (**one-sided test**) or

$H_a: \mu \neq 170$ (**two-sided test**)

TEST STATISTIC

This is an example of a one-sample test of a mean when σ is known.
Use this statistic to test the problem:

$$Z_{\text{stat}} = \frac{\bar{X} - \mu_0}{SE_{\bar{X}}} \quad \text{Follow standard normal distribution } N(0,1)$$

where $\mu_0 \equiv$ population mean assuming H_0 is true

$$\text{and } SE_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

ILLUSTRATIVE EXAMPLE: Z STATISTIC

For the illustrative example, $\mu_0 = 170$

We know $\sigma = 40$

Take an SRS of $n = 64$. Therefore

$$SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{40}{\sqrt{64}} = 5$$

If we found a sample mean of 173, then

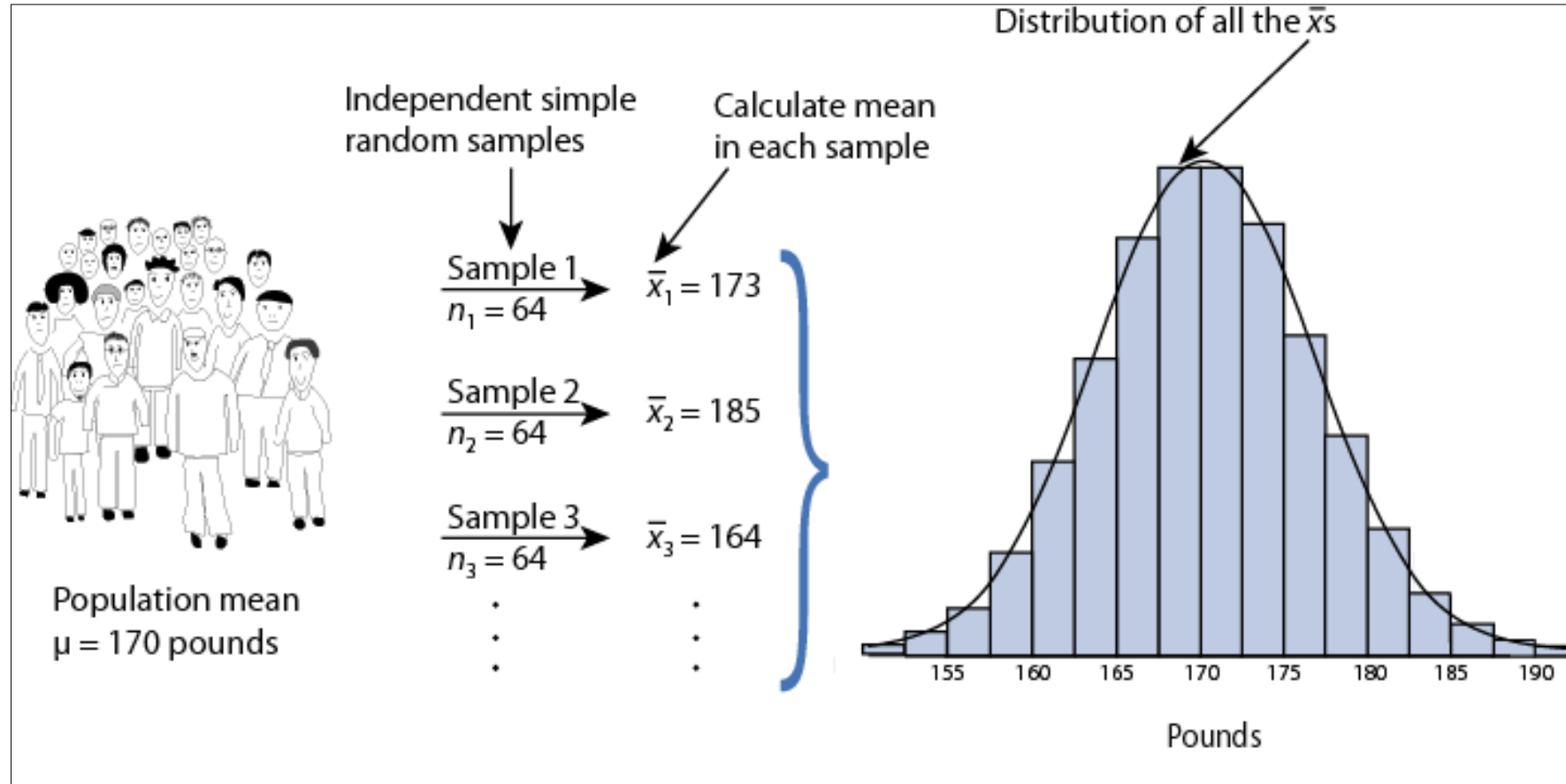
$$z_{\text{stat}} = \frac{\bar{x} - \mu_0}{SE_{\bar{x}}} = \frac{173 - 170}{5} = 0.60$$

ILLUSTRATIVE EXAMPLE: Z STATISTIC

If we found a sample mean of 185, then

$$z_{\text{stat}} = \frac{\bar{x} - \mu_0}{SE_{\bar{x}}} = \frac{185 - 170}{5} = 3.00$$

REASONING BEHIND Z_{STAT}



Sampling distribution of $\bar{x}$
under $H_0: \mu = 170$ for $n = 64 \Rightarrow$

$$\bar{x} \sim N(170, 5)$$

P-VALUE

The P -value answer the question: What is the probability of the observed test statistic or one more extreme **when H_0 is true?**

This corresponds to the AUC in the tail of the Standard Normal distribution beyond the z_{stat} .

Convert z statistics to P -value :

For $H_a: \mu > \mu_0 \Rightarrow P = \Pr(Z > z_{\text{stat}})$ = right-tail beyond z_{stat}

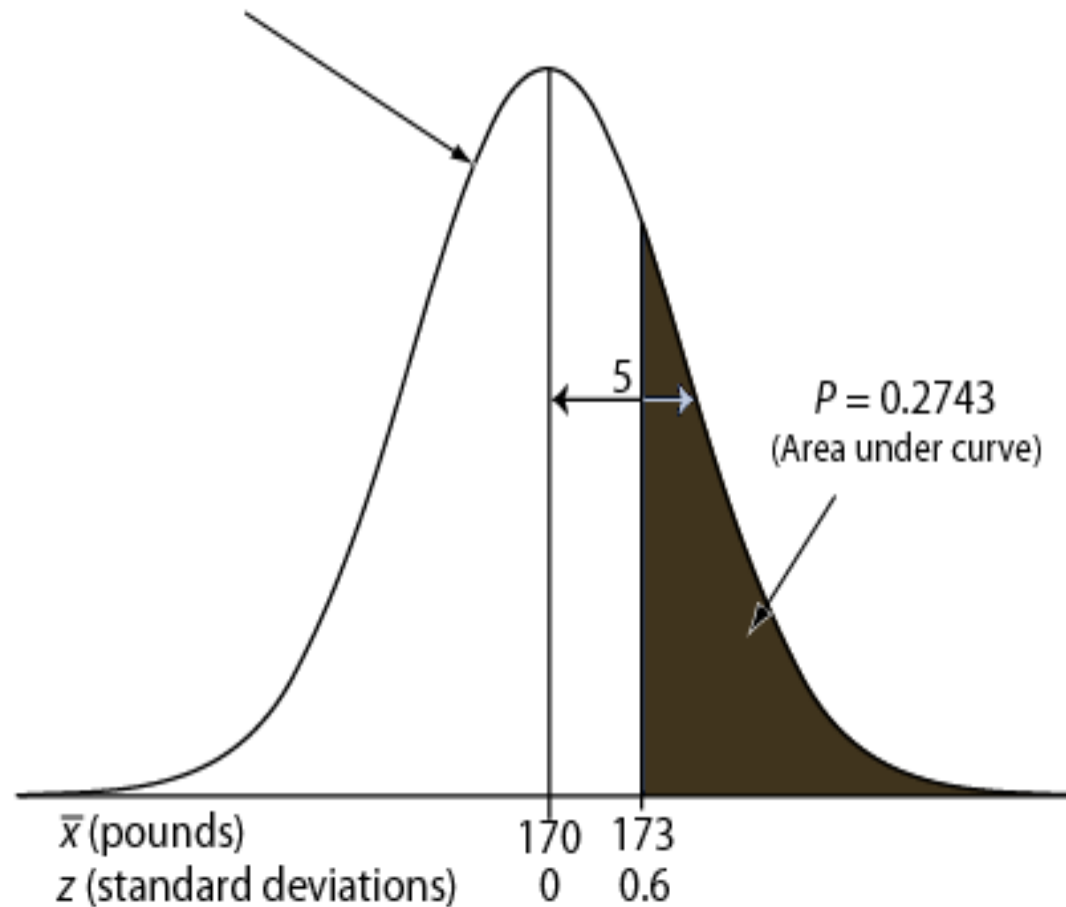
For $H_a: \mu < \mu_0 \Rightarrow P = \Pr(Z < z_{\text{stat}})$ = left tail beyond z_{stat}

For $H_a: \mu \neq \mu_0 \Rightarrow P = 2 \times \text{one-tailed } P\text{-value}$

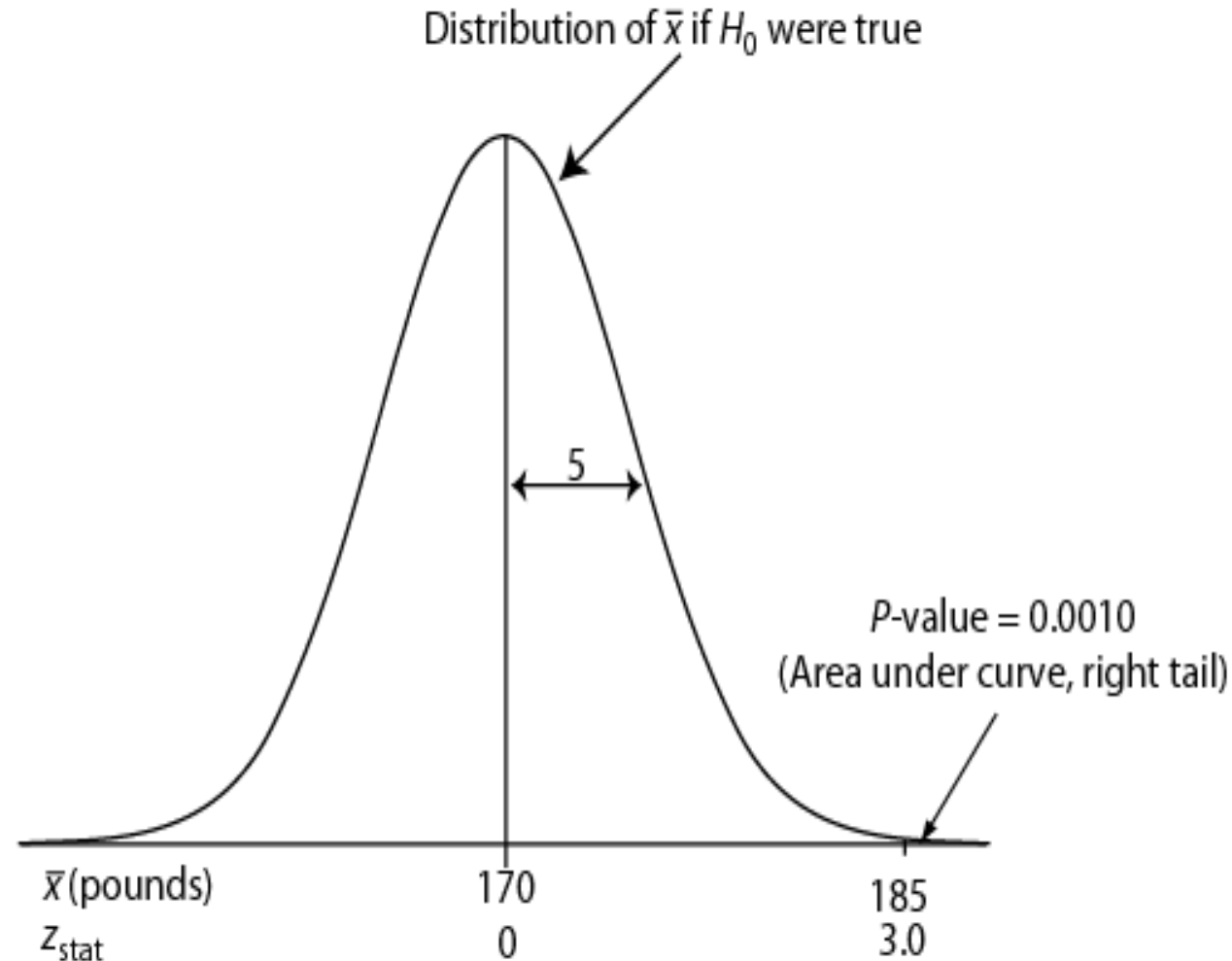
Use Tables or software to find these probabilities (next two slides).

ONE-SIDED P -VALUE FOR Z_{STAT} OF 0.6

Distribution of $\bar{x}$ and z_{stat} if H_0 were true



ONE-SIDED P -VALUE FOR Z_{STAT} OF 3.0

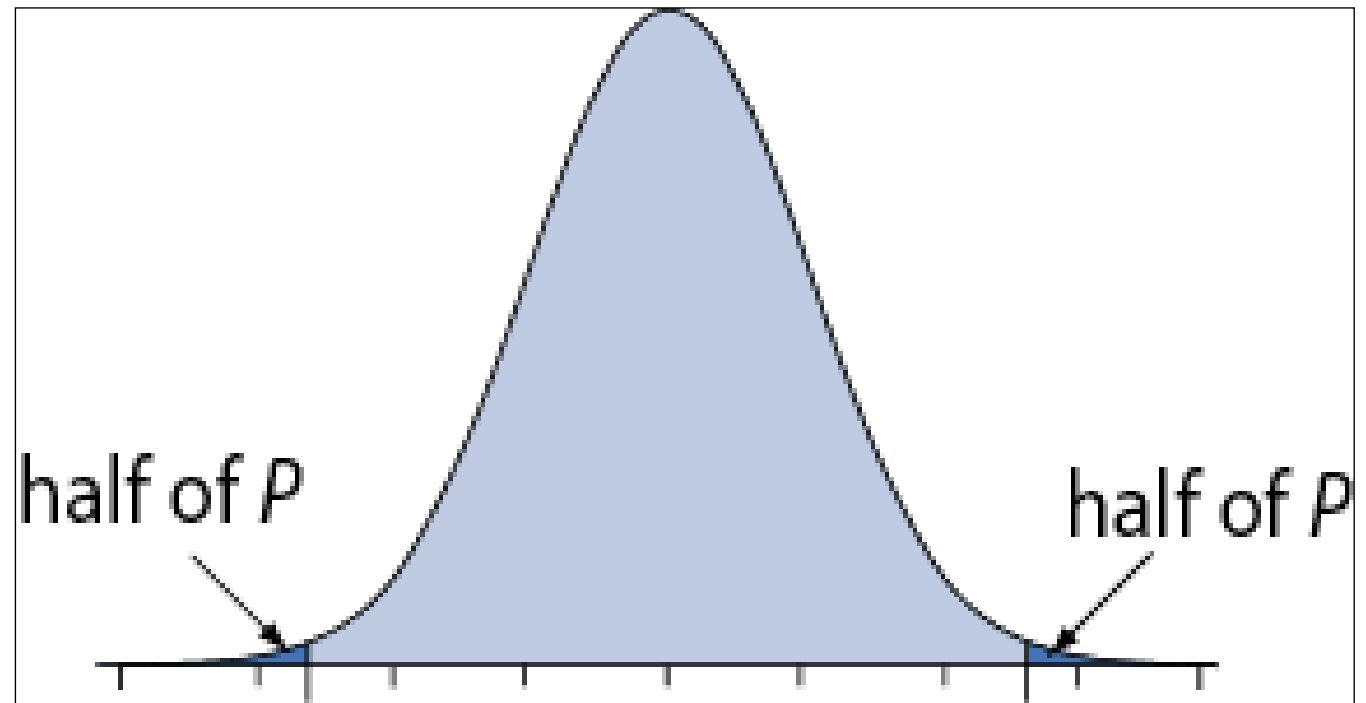


TWO-SIDED P -VALUE

One-sided $H_a \Rightarrow$ AUC in tail beyond z_{stat}

Two-sided $H_a \Rightarrow$ consider potential deviations in both directions $\Rightarrow$ double the one-sided P -value

Examples: If one-sided $P = 0.0010$, then two-sided $P = 2 \times 0.0010 = 0.0020$. If one-sided $P = 0.2743$, then two-sided $P = 2 \times 0.2743 = 0.5486$.



INTERPRETATION

P-value answer the question: What is the probability of the observed test statistic ... **when H_0 is true?**

Thus, smaller and smaller *P*-values provide stronger and stronger evidence against H_0

Small *P*-value $\Rightarrow$ strong evidence

INTERPRETATION

Conventions*

$P > 0.10 \Rightarrow$ non-significant evidence **against** H_0

$0.05 < P \leq 0.10 \Rightarrow$ marginally significant evidence

$0.01 < P \leq 0.05 \Rightarrow$ significant evidence against H_0

$P \leq 0.01 \Rightarrow$ highly significant evidence against H_0

Examples

$P = .27 \Rightarrow$ non-significant evidence against H_0

$P = .01 \Rightarrow$ highly significant evidence against H_0

*** It is *unwise* to draw firm borders for “significance”**

A-LEVEL (USED IN SOME SITUATIONS)

Let $\alpha \equiv$ probability of erroneously rejecting H_0

Set α threshold (e.g., let $\alpha = .10, .05$, or *whatever*)

Reject H_0 when $P \leq \alpha$

Retain H_0 when $P > \alpha$

Example: Set $\alpha = .10$. Find $P = 0.27 \Rightarrow$ retain H_0

Example: Set $\alpha = .01$. Find $P = .001 \Rightarrow$ reject H_0

(SUMMARY) ONE-SAMPLE Z TEST

Hypothesis statements

$H_0: \mu = \mu_0$ vs.

$H_a: \mu \neq \mu_0$ (two-sided) or

$H_a: \mu < \mu_0$ (left-sided) or

$H_a: \mu > \mu_0$ (right-sided)

Test statistic

P-value: convert z_{stat} to P value

Significance statement (usually not necessary)
$$z_{\text{stat}} = \frac{\bar{x} - \mu_0}{SE_{\bar{x}}} \text{ where } SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

CONDITIONS FOR Z TEST

σ known (not from data)

Population approximately Normal or large sample (central limit theorem)

SRS (Simple random Sample or facsimile)

Data valid

THE LAKE WOBEGON EXAMPLE

“WHERE ALL THE CHILDREN ARE ABOVE AVERAGE”

Let X represent Weschler **Adult Intelligence scores** (WAIS)

Typically, $X \sim N(100, 15)$

Take SRS of $n = 9$ from Lake Wobegon population

Data $\Rightarrow \{116, 128, 125, 119, 89, 99, 105, 116, 118\}$

Calculate: $\bar{x} = 112.8$

Does sample mean provide strong evidence that population mean $\mu > 100$?

EXAMPLE: “LAKE WOBEGON”

A. Hypotheses:

$H_0: \mu = 100$ versus

$H_a: \mu > 100$ (one-sided)

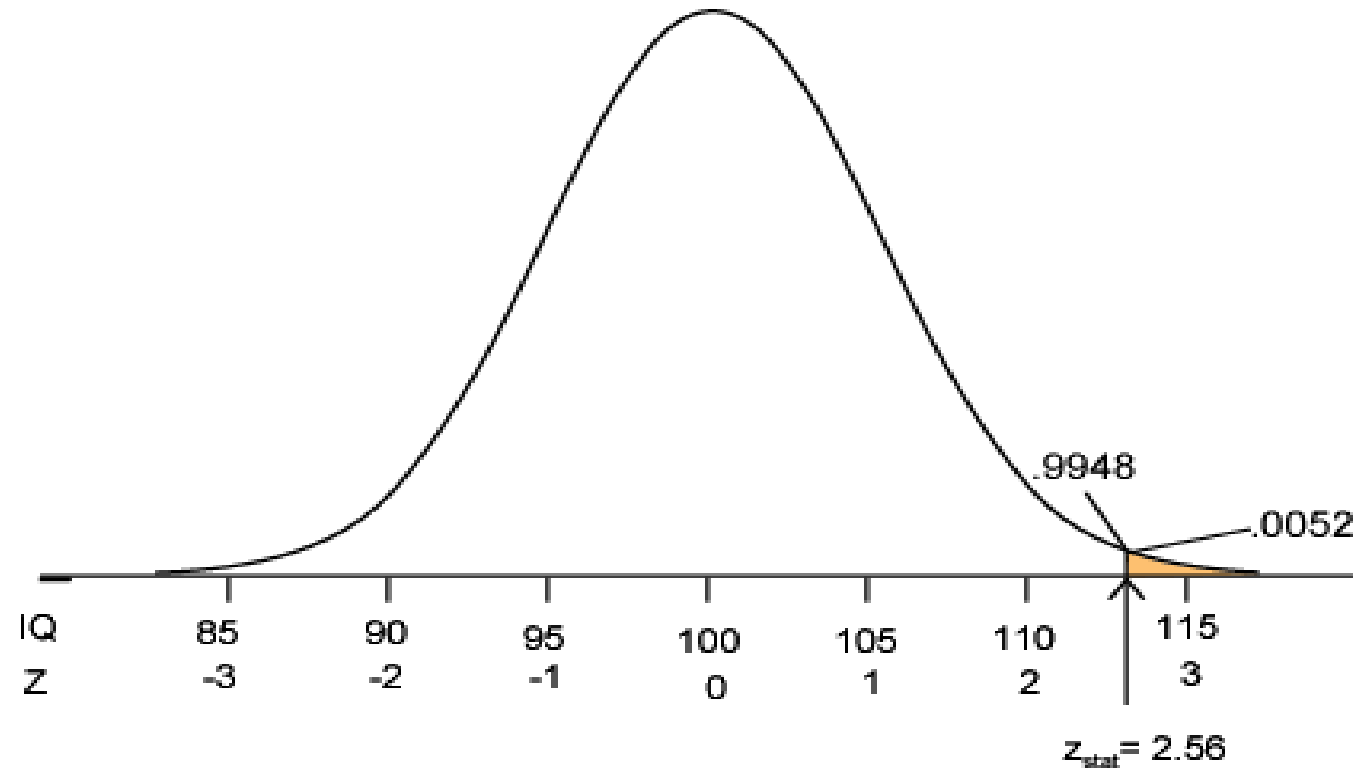
$H_a: \mu \neq 100$ (two-sided)

B. Test statistic:

$$SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{9}} = 5$$

$$z_{\text{stat}} = \frac{\bar{x} - \mu_0}{SE_{\bar{x}}} = \frac{112.8 - 100}{5} = 2.56$$

C. P-value: $P = \Pr(Z \geq 2.56) = 0.0052$



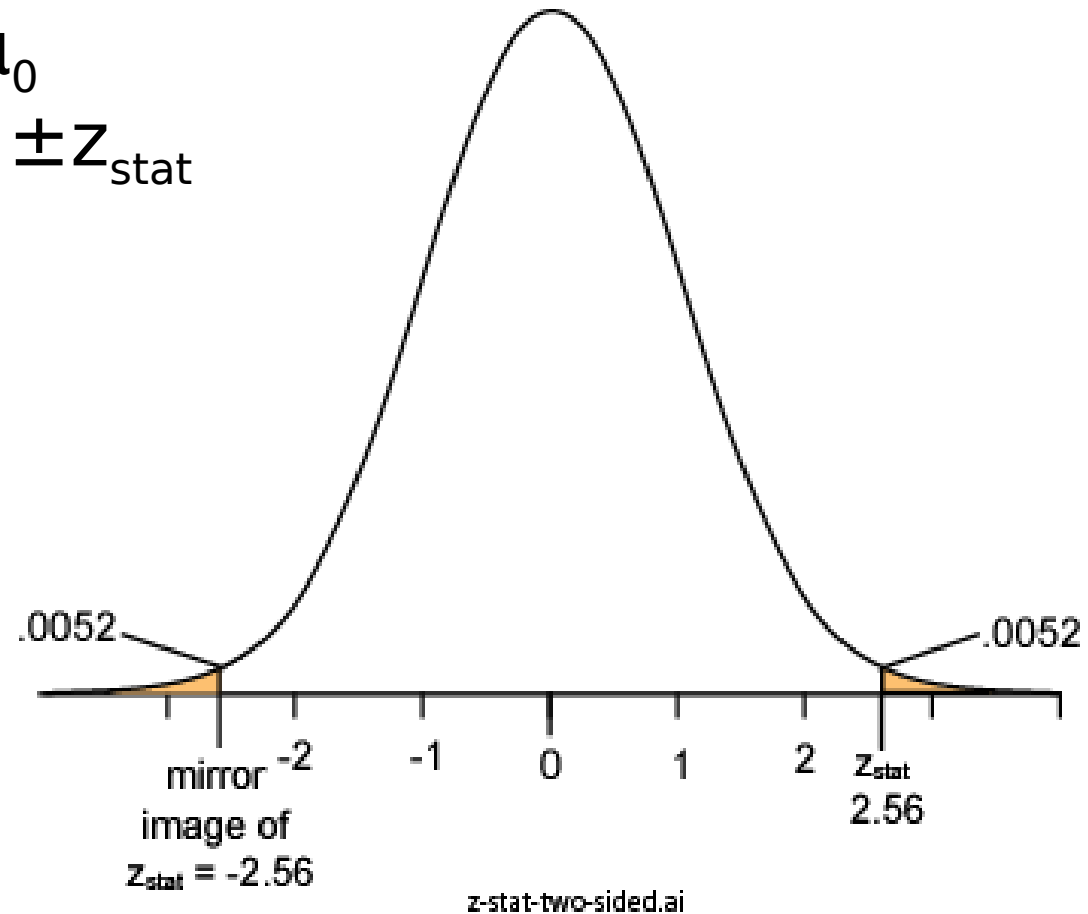
$P = .0052 \Rightarrow$ it is unlikely the sample came from this null distribution $\Rightarrow$ strong evidence against H_0

TWO-SIDED P -VALUE: LAKE WOBEGON

$$H_a: \mu \neq 100$$

Considers random deviations
“up” and “down” from μ_0
 $\Rightarrow$ tails above and below $\pm z_{\text{stat}}$

$$\begin{aligned}\text{Thus, two-sided } P \\ &= 2 \times 0.0052 \\ &= 0.0104\end{aligned}$$



POWER AND SAMPLE SIZE

Two types of decision errors:

Type I error = erroneous rejection of true H_0

Type II error = erroneous retention of false H_0

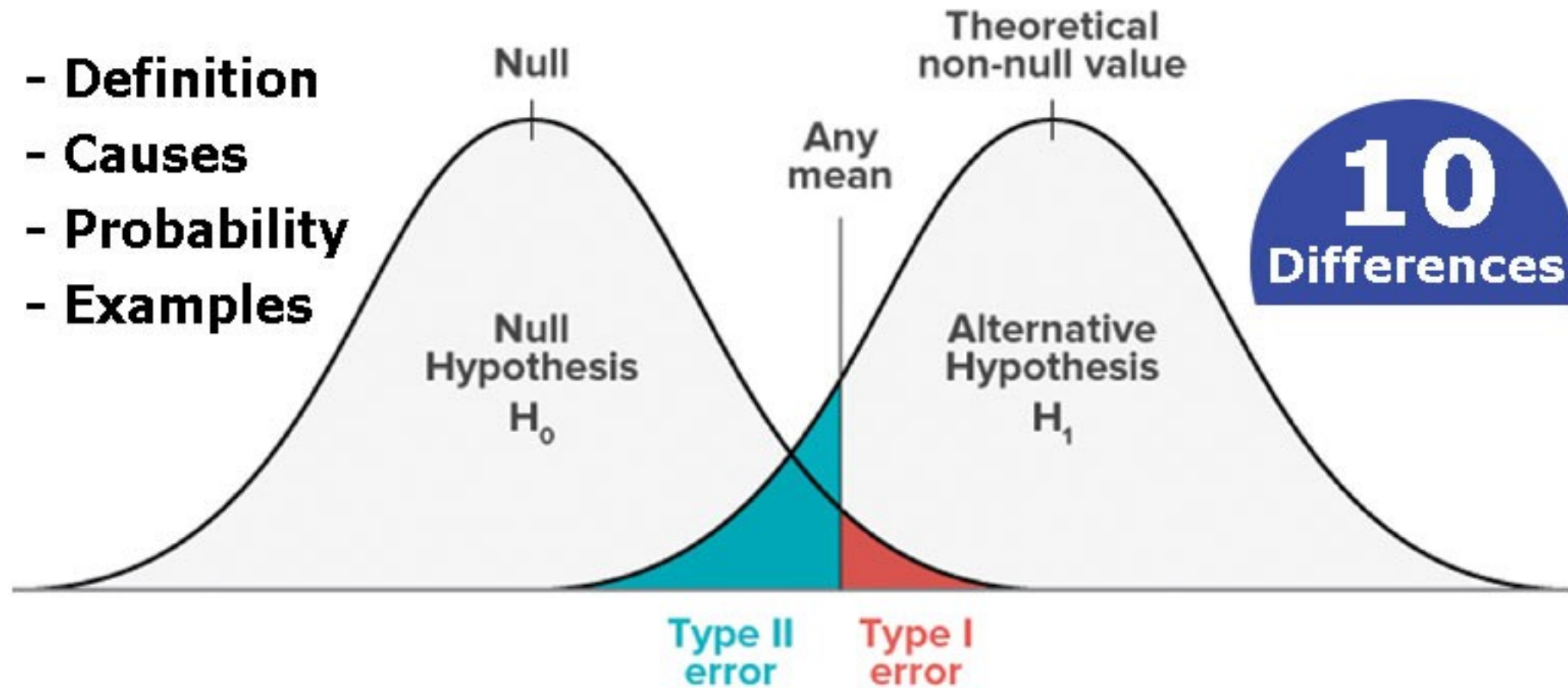
Decision	Truth	
	H_0 true	H_0 false
Retain H_0	Correct retention	Type II error
Reject H_0	Type I error	Correct rejection

$\alpha \equiv$ probability of a Type I error

$\beta \equiv$ Probability of a Type II error

Type I Error and Type II Error

- Definition
- Causes
- Probability
- Examples



POWER

$\beta \equiv$ probability of a Type II error

$\beta = \Pr(\text{retain } H_0 \mid H_0 \text{ false})$
(the “|” is read as “given”)

$1 - \beta = \text{“Power”} \equiv$ **probability of avoiding a Type II error**

$1 - \beta = \Pr(\text{reject } H_0 \mid H_0 \text{ false})$

POWER OF A Z TEST

$$1 - \beta = \Phi \left(-z_{1-\frac{\alpha}{2}} + \frac{|\mu_0 - \mu_a| \sqrt{n}}{\sigma} \right)$$

where

$\Phi(z)$ represent the cumulative probability of Standard Normal Z

μ_0 represent the population mean under the null hypothesis

μ_a represents the population mean under the alternative hypothesis

CALCULATING POWER: EXAMPLE

A study of $n = 16$ retains $H_0: \mu = 170$ at $\alpha = 0.05$ (two-sided); σ is 40. What was the power of test's conditions to identify a population mean of 190?

$$\begin{aligned} 1 - \beta &= \Phi \left(-z_{1-\frac{\alpha}{2}} + \frac{|\mu_0 - \mu_a| \sqrt{n}}{\sigma} \right) \\ &= \Phi \left(-1.96 + \frac{|170 - 190| \sqrt{16}}{40} \right) \\ &= \Phi(0.04) \\ &= 0.5160 \end{aligned}$$

REASONING BEHIND POWER

Competing sampling distributions

Top curve (next page) assumes H_0 is true

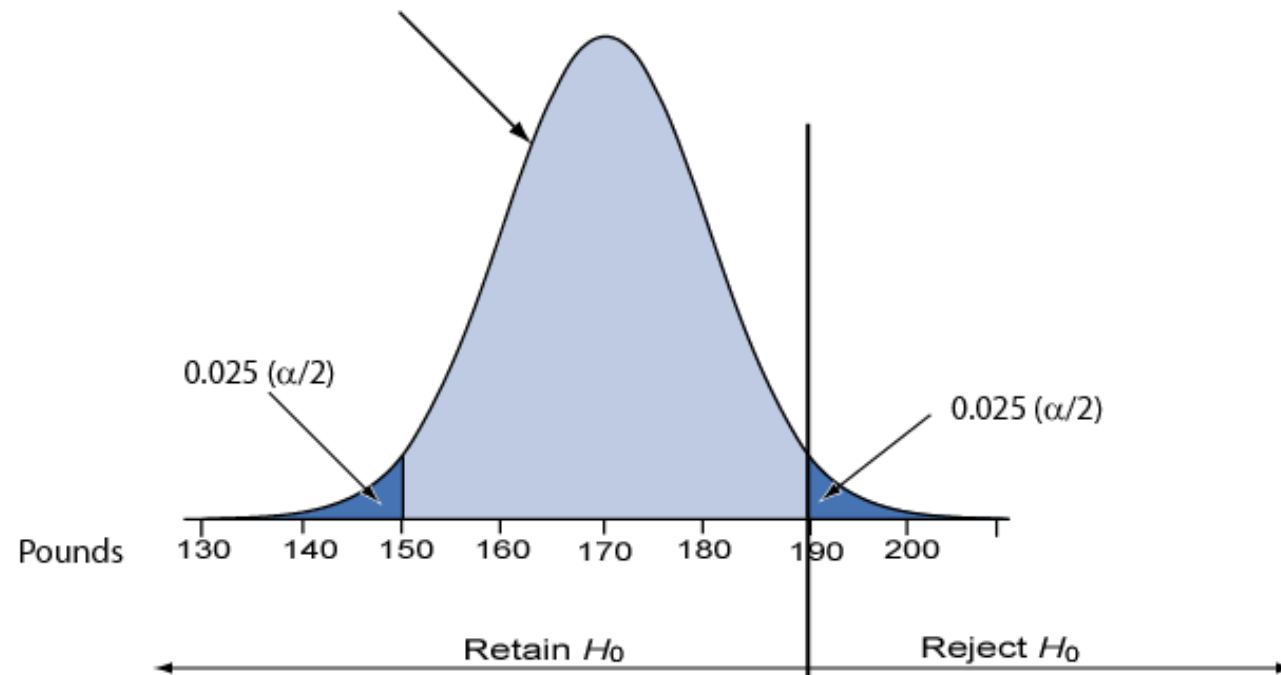
Bottom curve assumes H_a is true

α is set to 0.05 (two-sided)

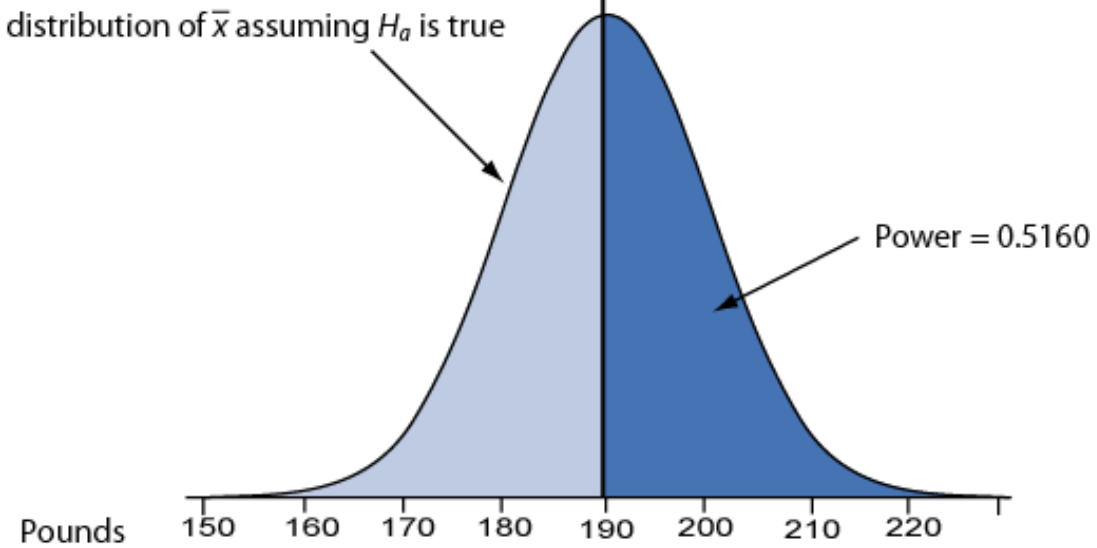
We will reject H_0 when a sample mean exceeds 189.6 (right tail, top curve)

The probability of getting a value greater than 189.6 on the bottom curve is 0.5160, corresponding to the power of the test

Sampling distribution of $\bar{x}$ assuming H_0 is true



Sampling distribution of $\bar{x}$ assuming H_a is true



SAMPLE SIZE REQUIREMENTS

Sample size for one-sample z test:

where

$$n = \frac{\sigma^2 \left(z_{1-\beta} + z_{1-\frac{\alpha}{2}} \right)^2}{\Delta^2}$$

$1 - \beta \equiv$ desired power

$\alpha \equiv$ desired significance level (two-sided)

$\sigma \equiv$ population standard deviation

$\Delta = \mu_0 - \mu_a \equiv$ the **difference worth detecting**

EXAMPLE: SAMPLE SIZE REQUIREMENT

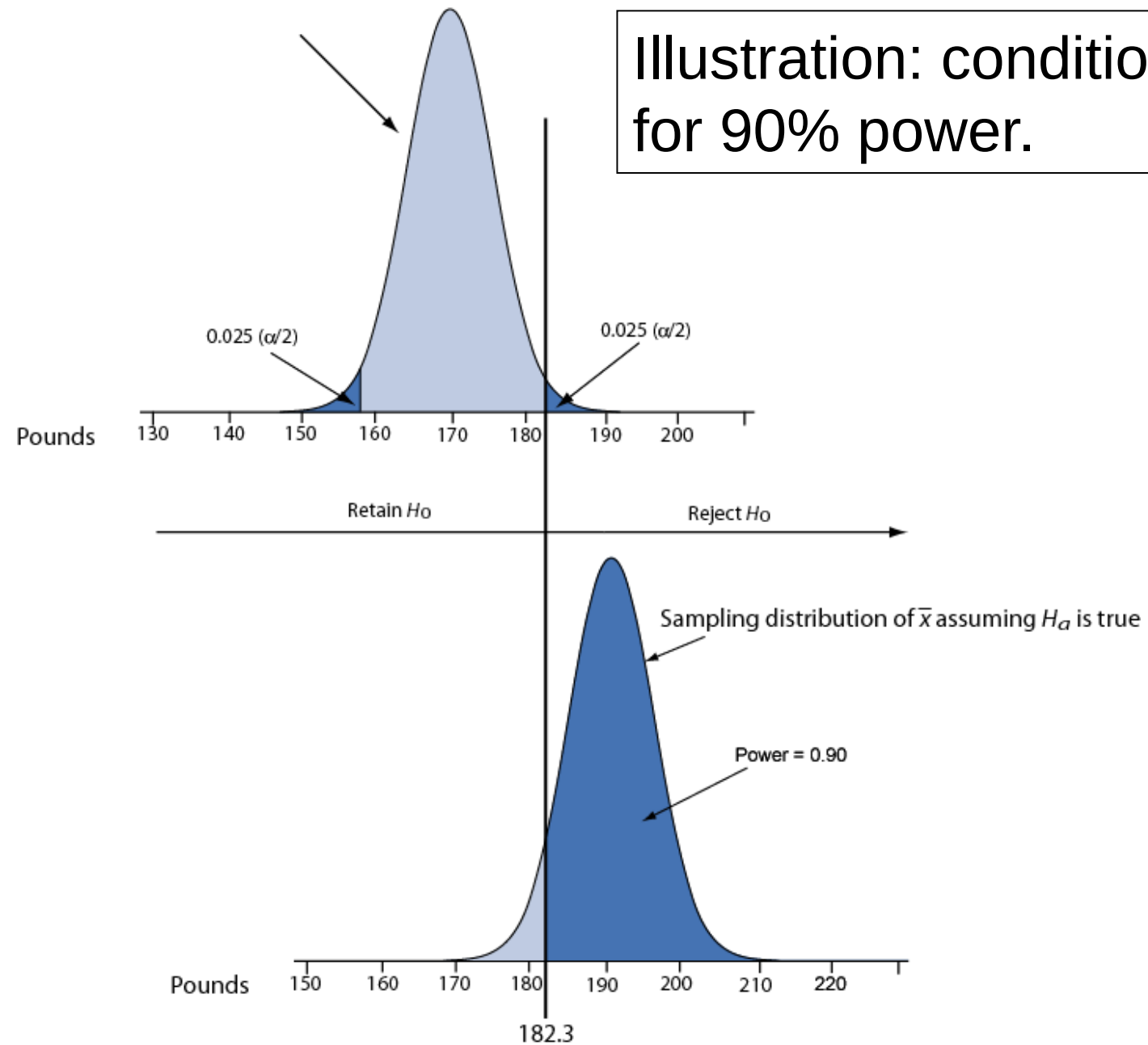
How large a sample is needed for a one-sample z test with 90% power and $\alpha = 0.05$ (two-tailed) when $\sigma = 40$?
Let $H_0: \mu = 170$ and $H_a: \mu = 190$ (thus, $\Delta = \mu_0 - \mu_a = 170 - 190 = -20$)

$$n = \frac{\sigma^2 \left(z_{1-\beta} + z_{1-\frac{\alpha}{2}} \right)^2}{\Delta^2} = \frac{40^2 (1.28 + 1.96)^2}{-20^2} = 41.99$$

Round up to 42 to ensure adequate power.

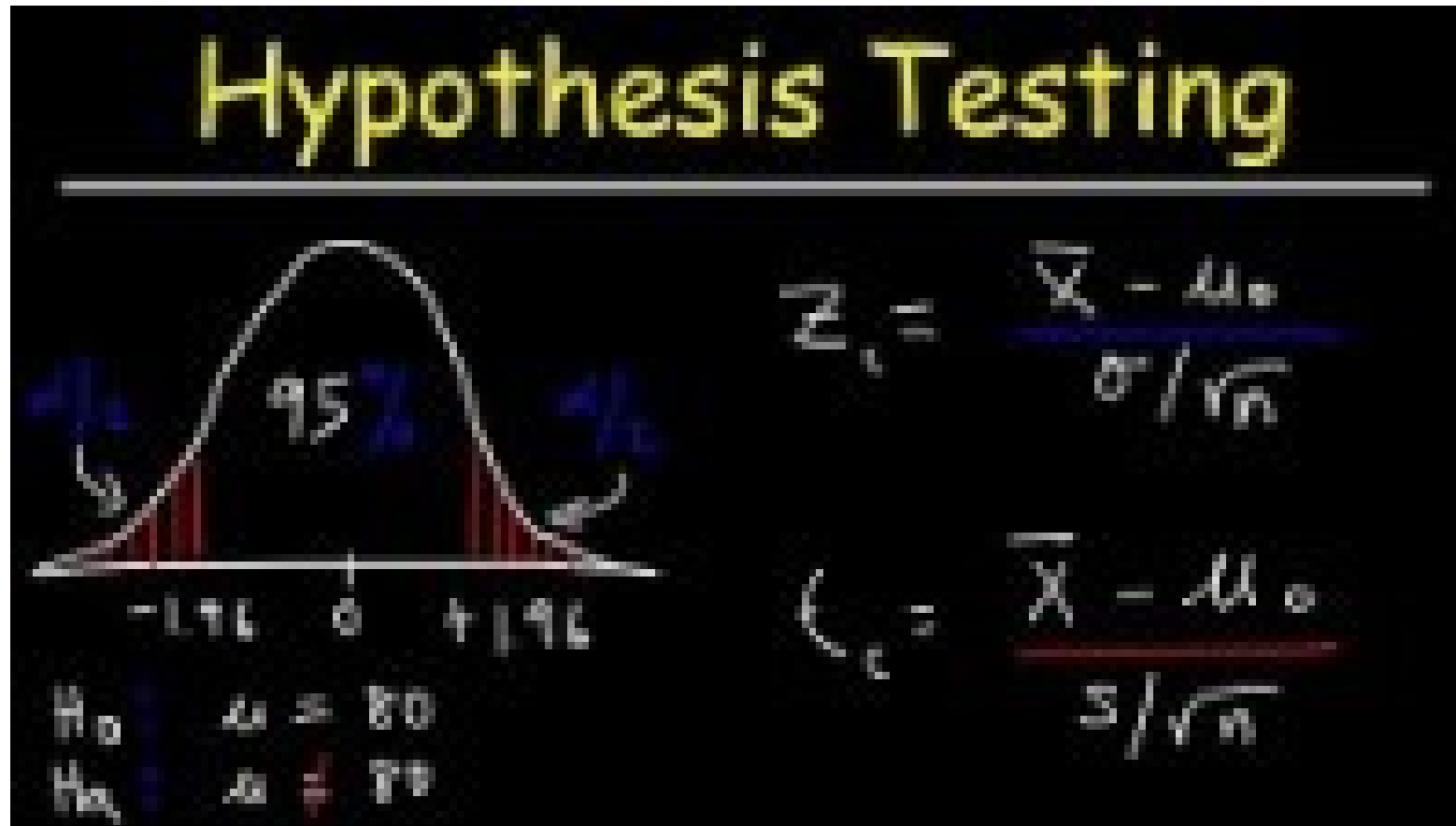
Sampling distribution of $\bar{x}$ assuming H_0 is true

Illustration: conditions
for 90% power.



ADDITIONAL RESOURCES

<https://www.simplilearn.com/tutorials/statistics-tutorial/hypothesis-testing-in-statistics>



https://www.youtube.com/watch?v=zJ8e_wAWUzE&t=191s

A neurologist is testing the effect of a drug on response time by injecting 100 rats with a unit dose of the drug, subjecting each to neurological stimulus, and recording its response time. The neurologist knows that the mean response time for rats not injected with the drug is 1.2 seconds. The mean of the 100 injected rats' response times is 1.05 seconds with a standard deviation of 0.2 seconds. Do you think that the drug has an effect?

Testing Hypothesis p-values

H_0 : Drug has no effect
 H_1 : Drug has an effect

Assume $\alpha = 0.05$
 $1.2 - 1.05 = 0.15$
 2.1

$$Z = \frac{1.05 - 1.2}{\frac{0.2}{\sqrt{100}}} = \frac{-0.15}{0.02} = -7.5$$
$$P(Z < -7.5) < 0.05$$



Khan Academy

<https://www.youtube.com/watch?v=-FtlH4svqx4>

INFERENCE ABOUT MEAN:

- Estimated Standard Error of the Mean
- Student's t Distribution
- One-Sample t Test
- Confidence Interval for μ
- Paired Samples
- Conditions for Inference
- Sample Size and Power

SIGMA NOT KNOWN

Prior chapter: σ was known before collecting data $\Rightarrow$ **z procedures** used to help infer μ

$$\mu = ?$$

$$\sigma = ?$$

When σ NOT known, calculate sample standard deviations **s** and use it to calculate this standard error:

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

ADDITIONAL UNCERTAINTY

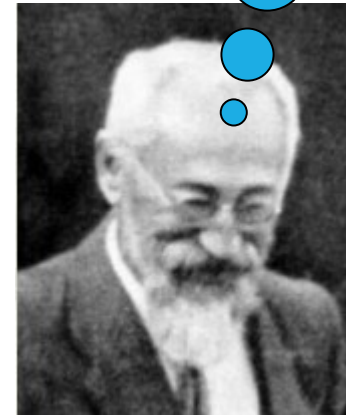
$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

Using s instead of σ adds uncertainty to inferences $\Rightarrow$ can NOT use z procedures

Instead, rely on **Student's t procedures**

The Normal distribution doesn't fit well



William Sealy
Gosset (1876–1937)

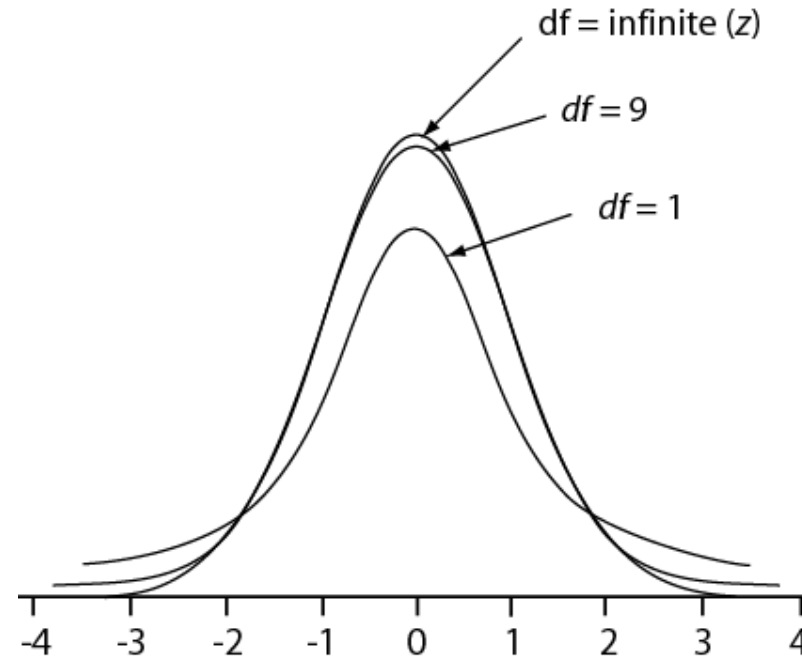
STUDENT'S T DISTRIBUTIONS

Family of probability distributions

Family members identified by degrees of freedom (df)

Similar to “ Z ”, but with broader tails

As df increases $\rightarrow$ tails get skinnier
 $\rightarrow t$ become like z



A t distribution with infinite degrees of freedom *is* a Standard Normal Z distribution

Constructing a 95% confidence interval (for the population mean)

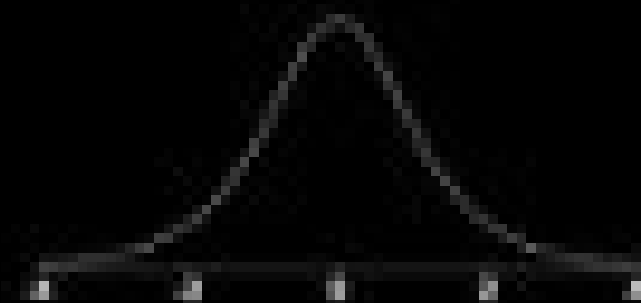
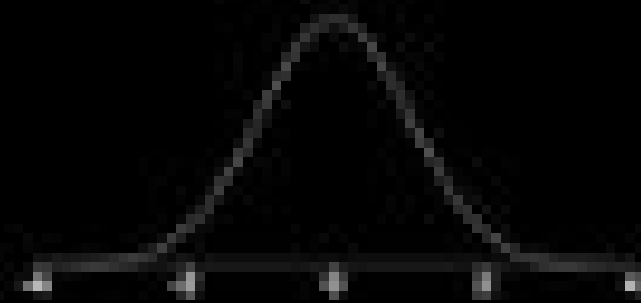


If σ is known:

$$\bar{X} \pm 1.96 \times \frac{\sigma}{\sqrt{n}}$$

If σ is not known:

$$\bar{X} \pm t \times \frac{s}{\sqrt{n}}$$



<https://www.youtube.com/watch?v=Uv6nGlgZMVw&t=388s>

TABLE C (*T* TABLE)

Cumulative probability		0.75	0.8	0.85	0.9	0.95	0.975	0.99	0.995	0.9975	0.999	0.9995
Upper tail probability		0.25	0.2	0.15	0.1	0.05	0.025	0.01	0.005	0.0025	0.001	0.0005
degrees of freedom	1	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	127.3	318.3	636.6
	2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	14.09	22.33	31.60
	3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	7.453	10.21	12.92
	4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.610
	5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	4.773	5.893	6.869
	6	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	4.317	5.208	5.959
	7	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.029	4.785	5.408
	8	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	3.833	4.501	5.041
	9	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	3.690	4.297	4.781
	10	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	3.581	4.144	4.587

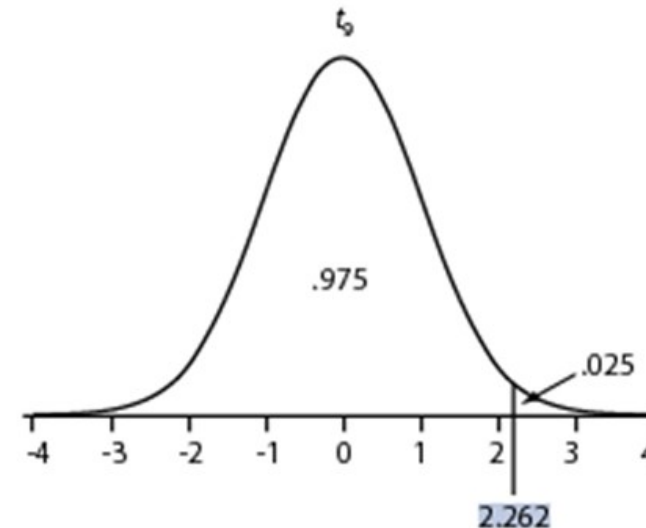
Rows $\Rightarrow df$

Columns $\Rightarrow$ probabilities

Entries $\Rightarrow t$ values

Notation: $t_{\text{cum_prob}, df} = t \text{ value}$

Example: $t_{.975, 9} = 2.262$



ONE-SAMPLE T TEST

Objective: test a claim about population mean μ

Conditions :

Simple Random Sample

Normal population *or* “large sample”

HYPOTHESIS STATEMENTS

Null hypothesis

$$H_0: \mu = \mu_0$$

where μ_0 represents the pop. mean expected by the null hypothesis

Alternative hypotheses

$$H_a: \mu < \mu_0 \text{ (one-sided, left)}$$

$$H_a: \mu > \mu_0 \text{ (one-sided, right)}$$

$$H_a: \mu \neq \mu_0 \text{ (two-sided)}$$



EXAMPLE

Do SIDS babies have lower average birth weights than a general population mean μ of 3300 gms?

$$H_0: \mu = 3300$$

$$H_a: \mu < 3300 \text{ (one-sided) or}$$

$$H_a: \mu \neq 3300 \text{ (two-sided)}$$



ONE-SAMPLE T TEST STATISTIC

$$t_{\text{stat}} = \frac{\bar{x} - \mu_0}{SE_{\bar{x}}}$$

where

$\bar{x} \equiv$ the sample mean

$\mu_0 \equiv$ expected population mean under H_0

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

This t statistic has $n - 1$ degrees of freedom

EXAMPLE (DATA)

SRS $n = 10$ birth weights (grams) of SIDS cases

2998

3740

2031

2804

2454

2780

2203

3803

3948

2144



$$\bar{x} = 2890.5 \text{ grams}$$

$$s = 720 \text{ grams}$$

$$n = 10$$

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}} = \frac{720}{\sqrt{10}} = 227.7 \text{ grams}$$

EXAMPLE

Testing $H_0: \mu = 3300$

$$t_{\text{stat}} = \frac{\bar{X} - \mu_0}{SE_{\bar{X}}} = \frac{2890.5 - 3300}{227.7} = -1.80$$

This statistic has

$$df = n - 1 = 10 - 1 = 9$$

P-VALUE VIA TABLE C

Bracket $|t_{\text{stat}}|$ between t critical values

For $|t_{\text{stat}}| = 1.80$ with 9 df

Table C.

Upper-tail P	0.25	0.20	0.15	0.10	0.05	0.025
$df = 9$	0.703	0.883	1.100	1.383	1.833	2.262

$|t_{\text{stat}}| = 1.80$

Thus $\Rightarrow$ One-tailed: $0.05 < P < 0.10$

Two-tailed: $0.10 < P < 0.20$

FOR A MORE PRECISE *P*-VALUE USE A COMPUTER UTILITY

StaTable for Java (Version 1.0.2)

Distribution:
Continuous Student's t

t -1.8

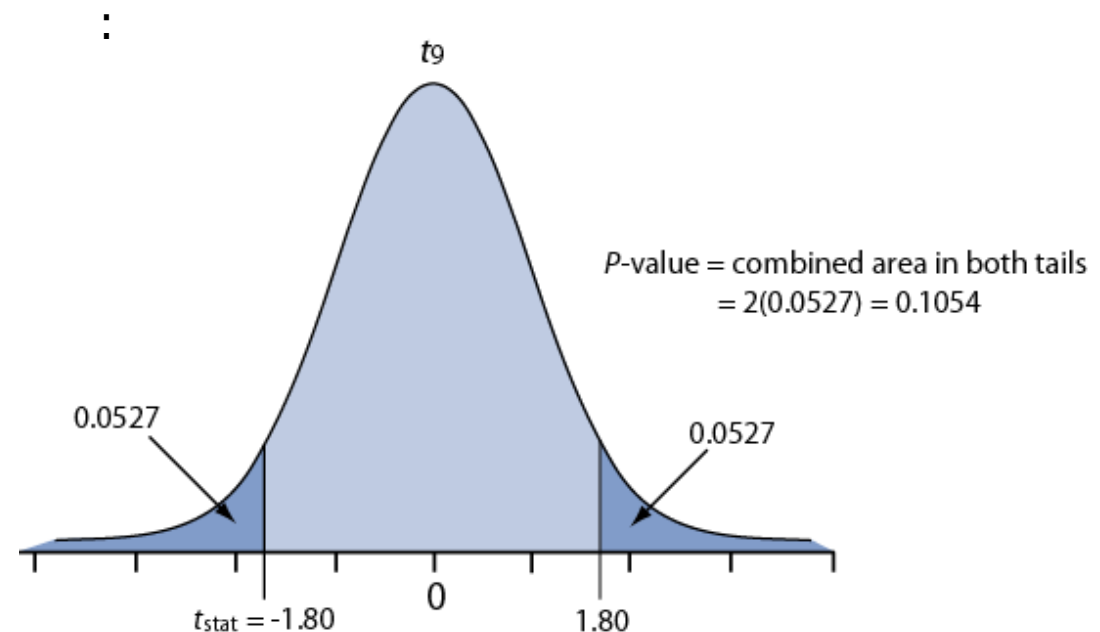
df 9.0

Left-tail 0.052695

Two-tail 0.105391

Here's output from the free utility
StaTable

Graphically



INTERPRETATION

Testing $H_0: \mu = 3300$ gms

Two-tailed $P > .10$

Conclude: weak evidence against H_0

The sample mean (2890.5) is NOT significantly different from 3300



$(1 - \alpha) * 100 - \textit{Confidence Interval}$

$$\bar{x} \pm t_{n-1, 1-\frac{\alpha}{2}} \cdot SE_{\bar{x}}$$

where

From
table

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

SAME DATA

$$\bar{x} = 2890.5 \quad s = 720 \quad n = 10$$

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}} = \frac{720}{\sqrt{10}} = 227.68$$

For 95% confidence use :

$$t_{10-1, 1-\frac{.05}{2}} = t_{9, .975} = 2.262$$

$$\begin{aligned} 95\% \text{ CI for } \mu &= \bar{x} \pm t_{9, .975} \cdot SE_{\bar{x}} = 2890.5 \pm (2.262)(227.68) \\ &= 2890.5 \pm 515.1 \\ &= (2375 \text{ to } 3406) \text{ grams} \end{aligned}$$

Interpretation: Population mean μ is between 2375 and 3406 grams with 95% confidence

TWO SAMPLE T TEST (GROUP DIFFERENCE)

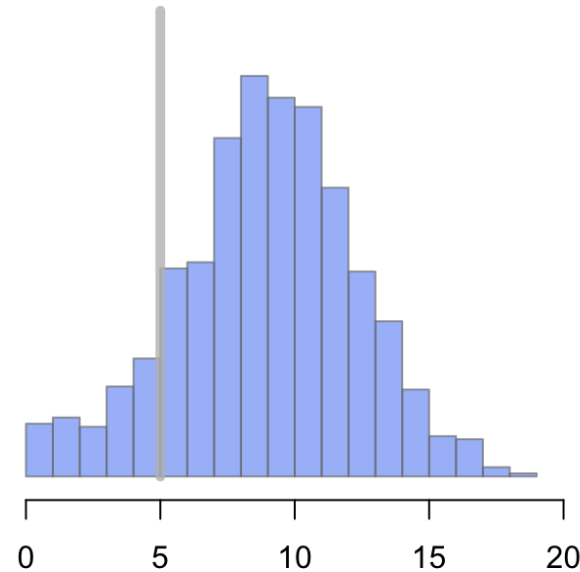
$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Unequal
variance

$$df = n_1 + n_2 - 2$$

1-Sample t-test

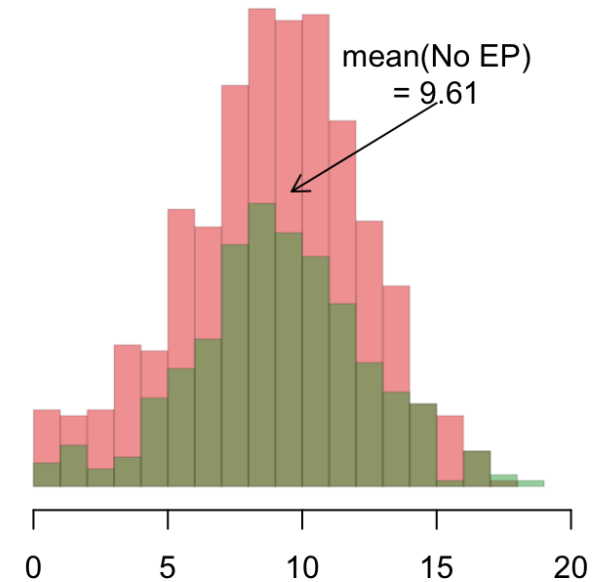
Null Hypothesis
Mean = 5



2-Sample t-test

mean(EP)
= 9.34

mean(No EP)
= 9.61



Kaito grows tomatoes in two separate fields. When the tomatoes are ready to be picked, he is curious as to whether the sizes of his tomato plants differ between the two fields. He takes a random sample of plants from each field and measures the heights of the plants. Here is a summary of the results:

$$\alpha = 0.05$$

$$H_0: \mu_A = \mu_B$$

Assume

$$H_a: \mu_A \neq \mu_B$$

	Field A	Field B
Mean	1.3 m	1.6 m
Standard deviation	0.5 m	0.3 m
Number of plants	22	24

$$t = \frac{\bar{X}_A - \bar{X}_B}{\sqrt{\frac{S_A^2}{n_A} + \frac{S_B^2}{n_B}}} = \frac{1.3 - 1.6}{\sqrt{\frac{0.25}{22} + \frac{0.09}{24}}} \approx -2.44$$

<https://www.youtube.com/watch?v=NkGvw18zIGQ>



Khan Academy

SEL study-1.0 (MH, 2022) : Relationship skills

Measurement of socio-emotional development among middle school children in Maharashtra

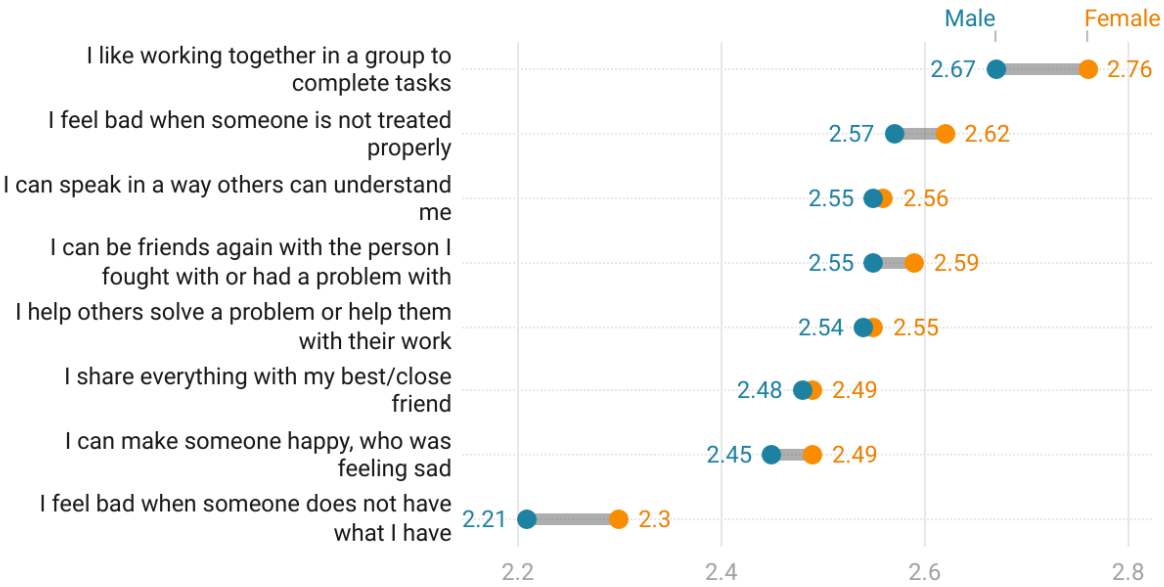


Chart: Shivakumar Jolad & Shalaka Shah • Source: Leadership for Equity • Created with Datawrapper

$$H_0 : \mu_M = \mu_W$$

Our research hypothesis is that the means are not equal, so we would say:

$$H_1 : \mu_M \neq \mu_W$$

SEL study-1.0: Self Management- MH, 2022

Measurement of socio-emotional development among middle school children in Maharashtra

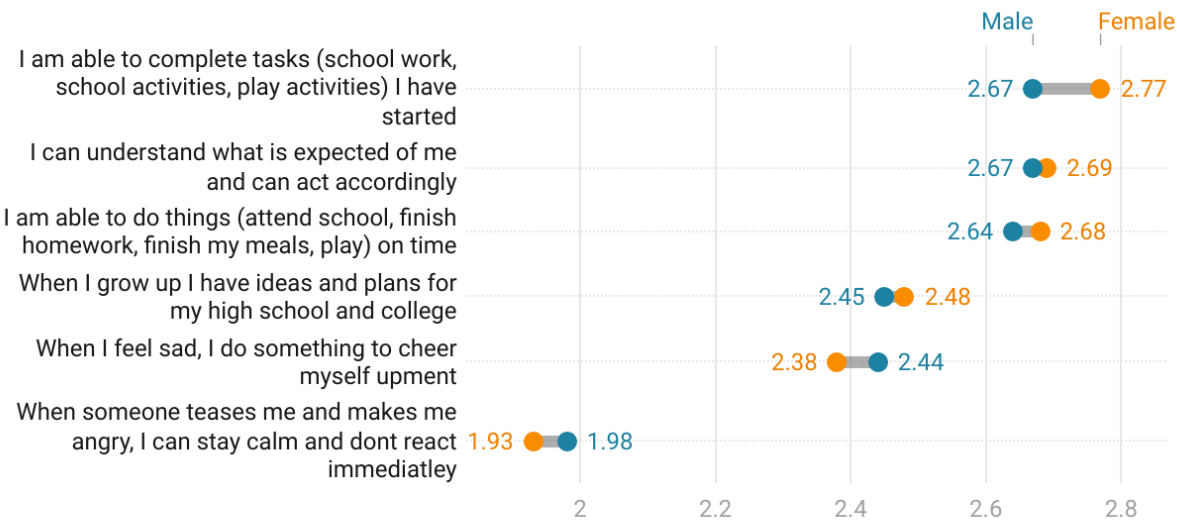


Chart: Shivakumar Jolad & Shalaka Shah • Source: Leadership for Equity • Created with Datawrapper

SEL study-1.0 (MH, 2022) : Social Awareness

Measurement of socio-emotional development among middle school children in Maharashtra

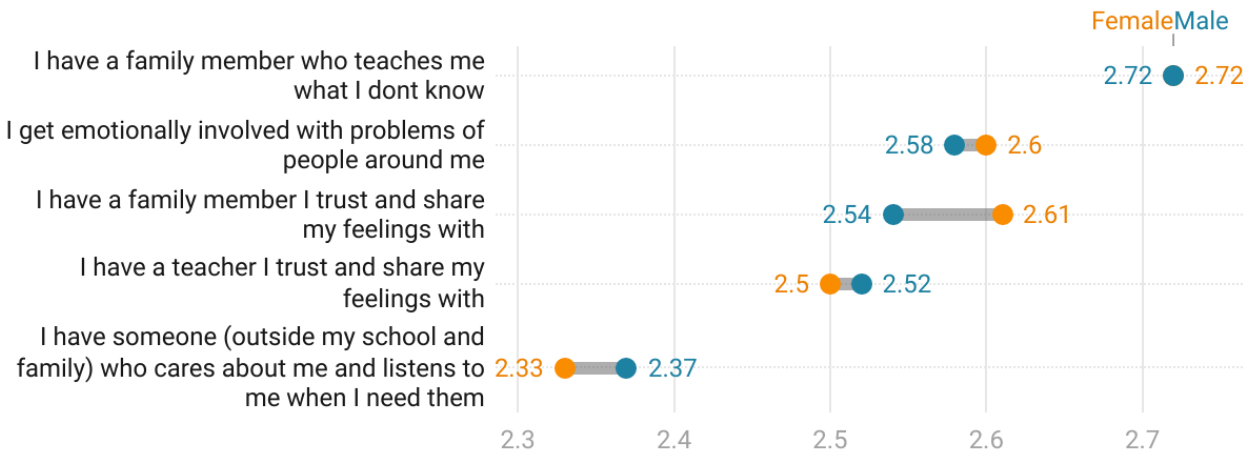


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SEL study-1.0 (MH, 2022) : Self Awareness

Measurement of socio-emotional development among middle school children in Maharashtra

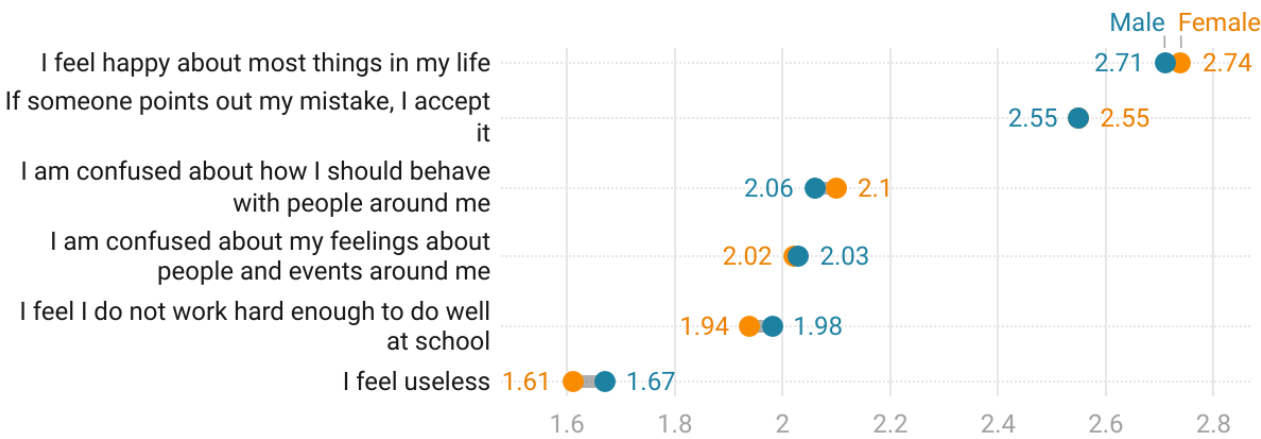


Chart: Shivakumar Jolad & Shalaka Shah • Source: Leadership for Equity • Created with Datawrapper

		Equality of Variances		t-test for Equality of Means				
						Significance		
SEL Domain		F	Sig.	t	df	One-Sided p	Two-Sided p	Mean Difference
	I can understand what is expected of me and can act accordingly	1.978	0.160	0.842	1846	0.200	0.400	0.020
	When I grow up I have ideas and plans for my high school and college	0.716	0.398	0.949	1851	0.171	0.343	0.029
	When someone teases me and makes me angry, I can stay calm and dont react immediatley	0.759	0.384	-1.521	1857	0.064	0.128	-0.045
	When I feel sad, I do something to cheer myself upment	3.449	0.063	-2.143	1861	0.016	0.032	-0.064
Self Regulation/ Self Management	I am able to complete tasks (school work, school activities, play activities) I have started	53.504	0.000	4.166	1864	0.000	0.000	0.096
Self discipline/Self Management	I am able to do things (attend school, finish homework, finish my	10.653	0.001	1.500	1869	0.067	0.134	0.037